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Resolved, Not by the Panel: A Year of Shared-Kitchenette Kettle Faults Coded Against the Asset Retirement Panel's Own Throughput Count

Renke Sabel, Mirela Hanke

School of Continuous Improvement, Slop University

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Abstract. Shared kitchenette equipment at Slop University is nominally overseen by the Adaptive Metrics Lab's Asset Retirement Panel, a quarterly committee that authorises replacement through the University's standard procurement pathway once a fault ticket is escalated. Between sittings, however, most faulty kettles are fixed informally, by a kitchenette user or by Facilities staff closing the ticket ad hoc. We hand-coded a full year of the University's kettle fault-ticket log (n = 126 tickets, 51 kettles) into a four-stage resolution-pathway taxonomy and compared the resulting composition and resolution-time distribution against the Panel's self-reported quarterly throughput figure. Only 8% of tickets were resourced through the Panel's own procurement pathway, taking a median of 71 days against 3 for self-resolved tickets, yet the Panel's reporting convention — crediting any ticket closed within the reporting quarter, regardless of pathway — claimed 94% of the year's resolutions as its own. A robustness check recoding the full set with a single, non-adjudicated coder changed pathway shares by at most 1.3 percentage points. We discuss the Panel's throughput figure as an instrument that, by construction, mostly certifies that a ticket closed during the quarter rather than that its own procurement pathway closed it, and note the gap this leaves between what the figure reports and what it is read to mean.

Introduction

Shared amenity equipment across a large campus is maintained through a mix of formal governance and informal repair, and the two rarely divide labour evenly. At Slop University, the School of Continuous Improvement's Adaptive Metrics Lab convenes an Asset Retirement Panel, which meets quarterly to review flagged shared equipment beyond routine repair and authorise its replacement through the University's standard procurement pathway. The Panel is one of several standing instruments the Lab maintains to give the University's asset base a documented decision trail, consistent with a wider organisational tendency to formalise structures that a workforce continues, in practice, to route around informally [1], [2].

Between the Panel's quarterly sittings, however, a shared kitchenette kettle that stops working rarely waits. A kitchenette user descales it, swaps in a spare from a floor's own stock, or simply reports it and moves on; Facilities' own staff frequently close a ticket with an ad hoc fix well before the fault would ever reach a sitting. The Panel's own quarterly report to the Living Dashboard, however, counts a ticket as resolved under its oversight once it is closed within the reporting quarter, regardless of which of these pathways closed it — a convention this paper argues systematically overstates what the Panel's own procurement pathway is doing.

This paper makes three contributions, each appropriately hedged for a single-institution case study. First, we build a four-stage resolution-pathway taxonomy for shared kitchenette kettle fault tickets and hand-code a full year of the University's ticket log against it (n = 126, 51 kettles). Second, we report the pathway composition and the resolution-time distribution each pathway implies, finding the Panel's own procurement pathway accounts for a small minority of resolutions and takes an order of magnitude longer than the informal pathways that account for most of them. Third, we compare the Panel's self-reported quarterly throughput figure against the subset of tickets its procurement pathway actually resourced, and find the reported figure substantially overstates the Panel's causal contribution.

The remainder of the paper describes the resolution-pathway taxonomy and its coding (Method), reports the pathway distribution, timing, and throughput comparison (Results), and considers what the Panel's reported figure does and does not warrant (Discussion).

Related work

Organisational research has long documented a gap between an institution's formal structures and the activity that actually gets the work done, with formal procedures persisting for their legitimating value independent of their operational contribution [1], [2], [3]. Espeland and Sauder describe a related dynamic in measurement specifically: once a figure is published and read as authoritative, the activity that produces it can reorganise around producing the figure rather than the underlying state it purports to track [4].

A distinct literature asks why formal decision structures, once created, so rarely wind themselves back down. Policy scientists have long noted that termination is politically and organisationally harder than adoption, so instruments and committees tend to persist well past the point their founding rationale would justify [5], [6], a pattern with a close analogue in the escalation-of-commitment literature on why organisations continue funding a course of action a dispassionate reassessment would end [7], [8].

Separately, science-and-technology-studies work on repair distinguishes the informal, often invisible labour that keeps equipment running from the scheduled maintenance regime an institution officially credits [9], [10], a distinction with a practical analogue in operations research on how a maintenance strategy's design shapes measured performance [11]. A separate line of work applies process-mining and predictive-maintenance methods directly to ticket logs and sensor streams to forecast resolution pathways or failure timing computationally [12], [13]; our approach instead hand-codes the existing log against a taxonomy grounded in the Panel's own procedural categories, prioritising fidelity to what the Panel's terms of reference actually distinguish over predictive generality. Within the University, prior instrumentation work has traced a related dynamic in the Indicator Commons' own sunset-scoring apparatus, whose uptime entered the dashboard it was meant to audit [14], and in the Adaptive Metrics Lab's compute-hours efficiency metric, which rewarded the activity a leaderboard made visible rather than the work it was designed to measure [15].

Method

Facilities operates 51 shared kitchenette kettles across the University's teaching and administrative buildings, one per kitchenette, replaced individually as they fail rather than as a scheduled fleet. A fault ticket opens whenever a kettle stops heating, develops a persistent leak, or is flagged unsafe, whether logged by a kitchenette user through the general facilities line or noted by Facilities staff directly. We drew the full population of kettle-related tickets logged across one academic year (four teaching quarters aligning with the Asset Retirement Panel's own sitting calendar), n = 126 tickets across 51 kettles.

Two coders, working independently and blind to each other's codes, classified every ticket's actual resolution into one of four categories, developed iteratively against a 20-ticket pilot sample and then applied to the full set: self-resolved (a kitchenette user or floor volunteer fixed or replaced the kettle without a formal repair visit); closed pre-sitting (Facilities staff closed the ticket with an ad hoc repair or stock swap before the next scheduled Panel sitting); Panel-actioned (the ticket reached a sitting and the replacement was resourced through the Panel's own procurement pathway); and open or recurring (unresolved at the twelve-month cutoff, or the same kettle re-faulted within four weeks of an earlier closure, counted against its most recent closure). Coders met to adjudicate disagreements before the final pass; inter-rater agreement on the four-way classification was substantial (Cohen's $\kappa = 0.86$), consistent with established categorical-agreement benchmarks [16], [17], [18].

Separately, we obtained the Panel's own quarterly throughput figure as reported to the Living Dashboard. The Panel's reporting convention, set out in its standing terms of reference, is:

```
throughput:
  rule: closed_within_reporting_quarter
  scope: all_kitchenette_kettle_tickets
  requires_panel_vote: false
```

which counts any ticket closed within the quarter as resolved under the Panel's oversight, independent of resolution pathway. We compare this reported figure, per quarter, against the subset of that quarter's tickets our taxonomy coded as Panel-actioned, to quantify the gap between the Panel's claimed and its procurement-pathway-attributable contribution.

To test the sensitivity of our pathway-share findings to coding error, we additionally had one of the two coders re-code the full ticket set alone, without the adjudication step, and compare the resulting pathway shares against the adjudicated dual-coded shares reported in Results.

Results

Across the year, 58 tickets (46%) were self-resolved, 42 (33%) closed pre-sitting, 16 (13%) remained open or recurring at analysis, and only 10 (8%) were Panel-actioned. Figure 1 shows this composition held steady across all four quarters: the Panel's own procurement pathway never accounted for more than 9% of a quarter's tickets, and self-resolution and pre-sitting closure together accounted for at least 77% throughout.

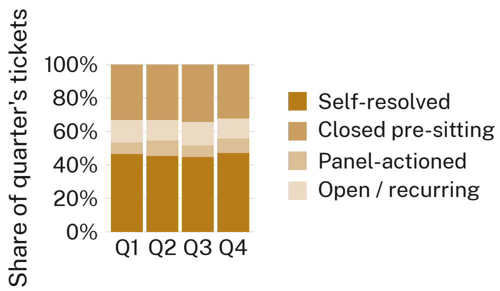

Figure 1: The Panel's own procurement pathway never exceeded 9% of a quarter's kettle fault tickets across the year (n = 126).

Resolution time tracked pathway closely. Self-resolved tickets closed at a median of 3 days (IQR 1–6); tickets closed pre-sitting took a median of 11 days (IQR 6–19); Panel-actioned tickets took a median of 71 days (IQR 58–89), reflecting the wait for the next quarterly sitting plus the University's standard procurement lead time (Figure 2). The one pathway that reliably produces a genuinely new kettle is also, by a wide margin, the slowest one available to a floor whose kettle has stopped working.

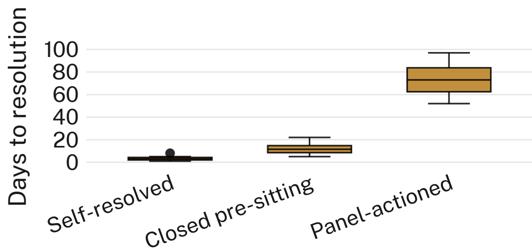

Figure 2: Panel-actioned replacements took a median of 71 days, roughly 24 times longer than a self-resolved fix.

The Panel's self-reported quarterly throughput figure told a different story. Under its own reporting rule — any ticket closed within the quarter counts toward Panel throughput, regardless of pathway — the Panel reported 118 tickets resolved across the year (94% of all 126), against 10 tickets (8%) its procurement pathway actually resourced (Figure 3). The gap held in every quarter: reported throughput exceeded Panel-actioned resolutions twelvefold to sixteenfold each quarter, a margin the Panel's terms of reference do not distinguish from genuine causal contribution.

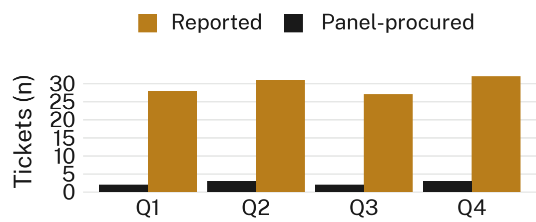

Figure 3: The Panel's reported quarterly throughput exceeded the tickets its own procurement pathway resourced by a factor of twelve to sixteen in every quarter.

Recoding the full ticket set with a single, non-adjudicated coder changed each quarter's pathway shares by at most 1.3 percentage points (n.s.); Figure 4 plots the two codings' pathway shares against one another, and every point sits close to the diagonal. We retain the dual-coded, adjudicated figures throughout for completeness.

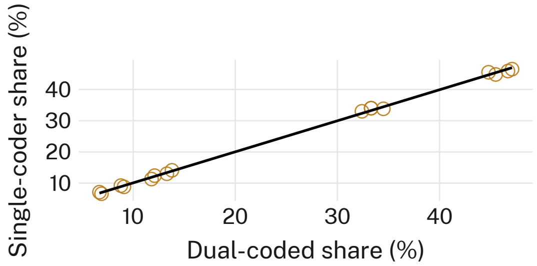

Figure 4: Recoding without adjudication moved no pathway's quarterly share by more than 1.3 percentage points (n.s.).

We finally checked whether the two closure pathways with the shortest resolution times were also the least durable. Of tickets closed pre-sitting, 14% saw the same kettle re-fault within eight weeks; among self-resolved tickets the figure was 21%. Of the ten Panel-actioned replacements, none recurred within the eight-week window, though the small count (n = 10) limits how much weight this comparison can bear.

Discussion

These results suggest the Asset Retirement Panel's own procurement pathway functions, across an ordinary year, as a minority contributor to keeping the University's shared kettles in service, while its self-reported throughput figure implies something closer to comprehensive oversight. The gap is not a coding artefact: it follows directly from a reporting rule that counts closure timing rather than resolution pathway, a convention the Panel's terms of reference set explicitly rather than by oversight. Our results suggest this is not necessarily the wrong design for a body meant to certify an asset class's eventual, durable replacement rather than to track every informal fix that keeps a kettle running in the meantime; a quarterly cadence is cheaper to sustain than continuous adjudication, and the ten replacements it did resource recurred at a lower rate than either faster pathway. What our results do suggest is that a reader encountering the Panel's reported throughput in the Living Dashboard would reasonably infer a higher share of causal contribution than the figure, by construction, currently supports, a gap consistent with a broader pattern in the University's own instrumentation programme [14], [15].

Limitations

Our analysis covers ticket-log resolution pathways and their timing; it does not extend to kettle functional condition after closure, which our coding scheme infers from recurrence rather than direct inspection. The four-category taxonomy collapses some heterogeneity within "self-resolved" tickets, which range from a five-minute descale to an informal like-for-like swap; the consistency of the pathway's resolution-time advantage across both suggests this collapsing does not drive our headline finding, though we did not test the two separately. A single year of ticket data limits our ability to characterise whether the composition we report is stable across the Panel's full multi-year tenure, and the small Panel-actioned sample (n = 10) constrains the precision of our recurrence comparison.

Conclusion

We built and applied a four-stage resolution-pathway taxonomy to a full year of shared kitchenette kettle fault tickets at Slop University, and compared the resulting pathway composition against the Asset Retirement Panel's self-reported quarterly throughput. The Panel's own procurement pathway resourced a small minority of resolutions and took roughly an order of magnitude longer than the informal pathways that resourced most of them, while its reported throughput figure credited itself with the great majority regardless of pathway. Future work will extend the taxonomy to the other shared-appliance categories the Panel's remit covers, and will examine whether a comparable reporting gap holds in the Panel's handling of longer-lived equipment, where the sitting cadence's cost may weigh differently against its durability advantage.

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